H1 and T1: a complete conditional lane-drop member

14 October 2026 — consolidated for the original VSL-B2-G1 confirmation gate

The user selected T1's finite physical terminal interpretation, fixed preparation handoff and restricted own-lane-first member. This note assembles the exact upstream and terminal laws for contract confirmation. It closes the bounded forward-model evidence review, not gradient or empirical qualification. S13 remains unchanged.

1. Geometry, intervention, information and units

A fixed road of length L has preparation length $L_p = L - b$ and an initially empty terminal length $b > 0$. An initial uniform footprint $[0, \ell]$ lies upstream, with $0 < \ell \le L_p$. There are no later external arrivals. The three densities are $e_0, a_0, b_0 \ge 0$, with $a_0 = b_0$ and

$$\rho_0 = e_0 + a_0 + b_0, \qquad M = \ell\rho_0 > 0, \qquad r = e_0 + a_0. \tag{1}$$

E is the ending lane, A the adjacent continuing lane, and B the other continuing lane. The scalar b_0 is a density; b is a length.

One constant common action $v \in [v_-, v_+]$, $0 < v_- < v_+$, sets preparation speed and the terminal free-flow parameter. Terminal queued motion need not equal v. All geometry, initial traffic, $w, k_j, h, \gamma, \alpha, R, R_B$ are fixed across actions and observers. Use the declared triangular family

$$k_c(v) = \frac{wk_j}{v + w}, \qquad C(v) = vk_c(v), \qquad w, k_j, h > 0. \tag{2}$$

The opportunity scale h is physical and fixed, not a numerical time step. Early and late work weights $\gamma, \alpha \ge 1$ are distinct parameters. $R, R_B > 0$ are fixed downstream work ceilings, not future receiving traces. Ordinary mass uses one normalized work unit.

Quantity	Meaning	Units
L, L_p, b, ℓ, h, z	lengths and physical progress	m
v, w	free-flow and backward-wave speeds	m/s
e, a, b_0, k_j, k_c	lane densities	veh-eq/m
$M, F_j, P_j, Y_j, N_j, q_j$	mass or cumulative mass	veh-eq
C, S, R, R_B	normalized receiving-work rates	work/s
f, g_E, u_j	transfer-equivalent or exit mass rates	veh-eq/s
$m = f/h$	lateral density source	veh-eq/(m s)
γ, α	work weights per transferred mass	dimensionless
TST	complete population-time integral	veh-eq s

2. H1: state to spare service to accepted early transfer

Initial densities satisfy $e_0, a_0, b_0 \le k_c(v_+)$. Inside the remaining homogeneous occupied body,

$$S(v,a) = \min\{C(v), w(k_j - a)\}, \quad f = \min\left\{ve, \frac{S(v,a) - va}{\gamma}\right\}, \quad \dot{e} = -\frac{f}{h}, \quad \dot{a} = \frac{f}{h}, \quad \dot{b}_0 = 0. \tag{3}$$

Own flow va claims receiving first; ve is the eager rate offer. f is an opportunity-equivalent bound for the same-coordinate lateral source, not a second longitudinal boundary flow. Actual transfer over $dx\,dt$ is $(f/h)dx\,dt$, not γ times that mass.

In the invariant free-flow branch $e + a = r$ and $S = C$, so

$$\dot{e} = -\frac{v}{h} \min\left\{e, \frac{k_c(v) - r + e}{\gamma}\right\}, \qquad a = r - e, \qquad b(t) = b_0. \tag{4}$$

At $e = 0$ transfer vanishes; at $a = k_c(v)$ spare service vanishes. The locally Lipschitz scalar equation therefore has a unique solution with

$$0 \le e(t) \le e_0, \qquad a_0 \le a(t) \le \min\{r, k_c(v)\}, \qquad b_0 \le k_c(v). \tag{5}$$

Uniform initial characteristics experience the same local source until they leave preparation. Consequently the remaining upstream body stays uniform and its edges move at v, provided the terminal entrance accepts it. No homogeneity is asserted inside the terminal.

With $z = vt$, $\delta(v) = k_c(v) - r$,

$$e_z = -\frac{1}{h} \min\{e, (\delta + e)/\gamma\}. \tag{6}$$

The explicit speed factor cancels. Speed-sensitive preparation must come through the declared normalized headroom, not elapsed time alone. Donor-limited and service-limited branch references from (z_s, e_s) are

$$e(z) = e_s e^{-(z-z_s)/h}, \qquad e(z) = (e_s + \delta)e^{-(z-z_s)/(\gamma h)} - \delta, \tag{7}$$

used only on their respective active branches.

3. The fixed handoff and conservative upstream balance

Early lateral sources stop at L_p. No further lateral source operates inside the terminal section; unfinished mass merges only at L. The initial terminal histories are zero. The preparation front and rear cross the handoff at

$$t_f^p = (L_p - \ell)/v, \qquad t_r^p = L_p/v.$$

Terminal entry rates are, almost everywhere,

$$\dot{F}_E = ve(t), \quad \dot{F}_A = va(t), \quad \dot{F}_B = vb_0 \quad \text{for } t_f^p < t < t_r^p, \tag{8}$$

and zero otherwise. Single instants at interval boundaries do not change the continuous-mass histories.

Let $s(t)$ be occupied length still in preparation: it equals ℓ before t_f^p, $L_p - vt$ during the exit interval, and zero after t_r^p. Stop the early source integral once $s = 0$ and put $J(t) = \int_0^t s(u)f(u)/h\,du$. Then

$$\begin{aligned} s\rho_0 + F_E + F_A + F_B &= M, \\ se + F_E + J &= \ell e_0, \\ sa + F_A - J &= \ell a_0, \qquad sb_0 + F_B = \ell b_0. \end{aligned} \tag{9}$$

These identities follow by differentiating on each interval and checking initial values. Already completed early transfers enter F_A once; they are not forced to merge again.

4. T1: physical storage, eligible demand and terminal receiving

Define delays $\tau = b/v$, $\tau_b = b/w$, and per-lane jam storage $H = k_j b$. Histories are extended as zero for negative times. For $j = E, A, B$,

$$P_j(t) = F_j(t - \tau), \quad N_j(t) = F_j(t) - Y_j(t), \quad q_j(t) = P_j(t) - Y_j(t). \tag{10}$$

P_j is cumulative minimum-travel eligibility, N_j physical occupied stock, and q_j eligible but unserved mass. The eligible queue is not the physical stock or the time when a traveler meets the queue's tail. Boundary counts retain the terminal LWR travel/storage meaning without independent interior position states.

Empty-link LTM receiving is at capacity while vacancy is positive:

$$V_j(t) = \begin{cases} k_j w t - F_j(t), & 0 < t \le \tau_b, \\ Y_j(t-\tau_b) + H - F_j(t), & t > \tau_b. \end{cases} \tag{11}$$

Use the conservative all-action domain

$$M < H, \qquad 0 \le \dot{F}_j(t) \le C(v). \tag{12}$$

After τ_b, $V_j \ge H - M > 0$. Before it, $F_j(t) \le C(v)t < k_j w t$. The entrance therefore never holds H1. This is a sufficient finite-storage proof, not an infinite buffer, a queue-only storage bound, or a general spillback solver.

At the ending-lane exit, demand is zero before τ. Afterwards it is $\dot{P}_E$ when $q_E = 0$ and $C(v)$ when $q_E > 0$. Both output histories obey minimum travel and finite sending. The present node is not claimed to be Jin's general invariant junction model.

5. The selected restricted compulsory server

The consolidated elementary member uses

$$v_+ r < R \le C(v_-), \qquad v_+ b_0 \le R_B \le C(v_-). \tag{13}$$

The upper ceilings are a conservative qualification refinement of K2: even the slowest free-flow lane can accommodate the declared exit budgets. The lower bound for R fits the entire ordinary $E + A$ load. Any terminal queue is therefore caused by extra forced work, not hidden ordinary overload.

Continuing traffic is served immediately: $Y_A = P_A$, $Y_B = P_B$, $u_A = \dot{P}_A$. Available ending service and actual flow are

$$c_E(t) = \min\left\{C(v), \frac{R - u_A(t)}{\alpha}\right\} > 0, \qquad g_E = \begin{cases} c_E(t), & q_E > 0, \\ \min\{\dot{P}_E, c_E(t)\}, & q_E = 0, \end{cases} \quad \dot{Y}_E = g_E. \tag{14}$$

Thus $\dot{q}_E = \dot{P}_E - g_E$, $q_E \ge 0$, $Y_E \le P_E$, and

$$u_A + \alpha g_E \le R, \qquad u_A + g_E \le R \le C(v). \tag{15}$$

Exactly $g_E dt$ exits, not $\alpha g_E dt$. There is no ending-lane forward escape, hidden second queue, lost residual mass or terminal re-transfer. On (13), the sending minimum in (14) is nonbinding because $(R - u_A)/\alpha \le R \le C(v_-) \le C(v)$. It is retained to show the physical sending interface.

Well-posedness of this member. The H1 solution fixes the absolutely continuous F_j and P_j. Let $p = P_E$, $K(t) = \int_0^t c_E(u)\,du$, and $X(t) = p(t) - K(t)$. The unique reflected fluid-queue solution is

$$q_E(t) = X(t) - \min_{0 \le u \le t} X(u), \qquad Y_E(t) = p(t) - q_E(t). \tag{16}$$

$X(0) = 0$; the running minimum prevents service of unavailable mass. Differentiating almost everywhere gives (14). This proves existence and uniqueness for the restricted boundary-coupled member, without invoking a theorem for arbitrary weighted network nodes.

All mass is eligible by $t_* = L/v$. Thereafter $u_A = 0$ and $c_E = R/\alpha > 0$, so

$$T_{\text{bound}} = \frac{L}{v} + \frac{\alpha q_E(L/v)}{R}. \tag{17}$$

This equals full clearance when a positive continuing tail reaches L/v, or when $q_E(L/v) > 0$; otherwise it is an upper bound. Using the bound in population-time integrals merely adds zero after the actual last exit.

6. One joint domain with active early service and positive forced waiting

The joint construction need not make early receiving irrelevant. The following scaled algebraic family demonstrates compatibility; it is not a calibrated scenario, selected traffic parameter set or rollout. Take arbitrary reference speed $u > 0$ and density $\kappa > 0$, and set

$$\begin{gathered} v_- = u, \quad v_+ = 2u, \quad w = 2u, \quad k_j = 3\kappa, \\ e_0 = \frac{7}{10}\kappa, \quad a_0 = b_0 = \frac{1}{4}\kappa, \quad \gamma = \alpha = 2, \quad R = \frac{39}{20}\kappa u, \quad R_B = \kappa u, \\ b = \ell > 0, \quad L_p = \ell + h\log 2, \quad L = L_p + b, \quad h > 0. \end{gathered} \tag{18}$$

Then $k_c(v_+) = 3\kappa/2$, $C(v_-) = 2\kappa u$, $C(v_+) = 3\kappa u$, $r = 19\kappa/20$, and

$$M = \frac{6}{5}\kappa\ell < 3\kappa\ell = H, \qquad v_+ r = \frac{19}{10}\kappa u < R < C(v_-), \qquad v_+ b_0 = \frac{1}{2}\kappa u < R_B < C(v_-). \tag{19}$$

Every initial density meets (5); invariant lane densities bound every entry rate by $C(v)$ over the whole action interval.

At the faster action, the initial normalized accepted early rate is

$$\min\left\{\frac{7}{10}, \frac{3/2 - 1/4}{2}\right\}\kappa = \frac{5}{8}\kappa < \frac{7}{10}\kappa. \tag{20}$$

Hence adjacent receiving limits the eager request. At the slower action the service alternative is $7\kappa/8$, so donor availability binds. The actions have different active preparation branches.

At the first terminal entry, exposure is $z_f = h\log 2$. The donor bound gives $e_f \geq e_0 e^{-z_f/h} = 7\kappa/20$. For v_+, the initial service branch switches at $e = 11\kappa/20$ after $z_s = 2h\log(25/22) < h\log 2$. Equation (7) gives the exact front residuals

$$e_f(v_-) = \frac{7}{20}\kappa, \qquad e_f(v_+) = \frac{11}{20}\kappa e^{-[h\log 2 - 2h\log(25/22)]/h} = \frac{125}{352}\kappa > \frac{7}{20}\kappa. \tag{21}$$

This is genuine, if modest, preparation dependence, not a speed-independent progress law.

The first fast eligible pulse has weighted demand

$$v_+(a_f + \alpha e_f) = 2u(r + e_f) \geq \frac{13}{5}\kappa u > R. \tag{22}$$

Positive queue growth therefore persists over a nonzero interval by continuity. Ordinary traffic still fits R, and (19) keeps terminal receiving nonbinding at the upstream interface. Active early limitation, positive forced waiting and protected homogeneous preparation coexist. No full-TST preference, interior optimum or observer collision is inferred.

7. Complete conservation and the population-time objective

For all times, with $s = 0$ after preparation empties,

$$s(t)\rho_0 + \sum_j N_j(t) + \sum_j Y_j(t) = M. \tag{23}$$

This follows from (9) and (10); work weights do not enter population counts. All mass exits by (17). Complete TST and mean time are

$$\mathrm{TST}(v) = \int_0^{T_{\mathrm{bound}}} \left[M - \sum_j Y_j(t) \right] dt, \qquad \overline{T} = \mathrm{TST}/M. \tag{24}$$

Every initial position $x \in [0, \ell]$ has no-delay exit time $(L - x)/v$. Only the ending stream has excess terminal delay. Since $\sum_j P_j$ counts all no-delay eligibility,

$$M - \sum_j Y_j = M - \sum_j P_j + q_E, \qquad \mathrm{TST} = M\frac{L - \ell/2}{v} + \int_0^{T_{\mathrm{bound}}} q_E(t)\, dt. \tag{25}$$

This identity includes both sections' travel and every unit's excess waiting. Eligible queue area measures excess time without equating q_E with spatial occupancy or moving the physical queue beyond the road.

Deletion check. For $\alpha = 1$, (13) admits all combined ordinary arrival flow. Thus $q_E = 0$, $Y_j = P_j$, and (25) gives the common fastest optimum. The same limit holds when the weighted arrivals already fit the node. This is a declared limiting reduction, not general aggregate equivalence.

Terminal-only reference. Ordinary and ending masses four each, uniformly eligible over two seconds, with $R = 4$, $\alpha = 2$ and slack sending, give ending queue peak two, drain time one, queue area three and complete-time sum eleven. The same masses eligible over four seconds give zero queue and complete-time sum sixteen. These are isolated exact references reused from K2, not H1 traffic experiments.

8. What information and speed sensitivity can be claimed

Known ℓ, initial symmetry and empty terminal histories give

$$e_0 = E_0/\ell, \qquad a_0 = b_0 = (M - E_0)/(2\ell). \tag{26}$$

Equations (4), (8), (10), (14) then determine the entire forward model. Aggregate mass plus E_0 is sufficient. Queues and boundary histories are predicted internal states, not additional initial measurements or future service oracles. No discrete order is needed among this homogeneous single-destination forced class.

For unchanged preparation labels per initial position, fixed R, α and the nonbinding-sending member, faster arrival cannot worsen complete TST. Put $W = P_A + \alpha P_E$ and $Z = Y_A + \alpha Y_E$. The work-conserving server gives

$$Z(t) = \inf_{0 \le u \le t} \{W(u) + R(t - u)\}, \qquad Y_A + Y_E + Y_B = Z/\alpha + (1 - 1/\alpha)P_A + P_B. \tag{27}$$

Earlier cumulative arrival makes W, P_A, P_B pointwise no smaller, hence departures pointwise no smaller. In the full H1 model preparation may change with advice, so this comparison does not settle its optimum. An interior optimum is not guaranteed by queueing.

The equations are piecewise smooth with preparation branch switches, action-dependent arrival endpoints, queue onset/emptying and capacity/domain boundaries. This note qualifies neither differentiated event handling nor autograd, finite differences, independently useful directions or a new information/action certificate. Those are original Pass 3 obligations.

9. Assumption register, provenance and gate disposition

- Retained task restrictions: finite cohort, initial continuing symmetry, one common fixed advice, ideal admitted free-flow response, continuous mass, fixed topology, full clearance and complete TST.
- H1 adaptations: a uniform occupied footprint, a uniform lateral source at the same road coordinate, an eager offer tied to physical progress, scale h, the selected triangular speed family and unrestricted free-flow admission to the terminal.
- T1 selected meaning: finite terminal reach, initially empty histories, no internal lateral sources after the handoff, point compulsory merging, strict own-lane-first priority and one delayed class.
- Declared terminal adaptations: constant R, R_B, α, conservative storage and ordinary-load restrictions, and all-action receiving ceilings in (13). None is a field-calibrated zipper or synchronized-flow law.

Carey, Balijepalli and Watling (2013), *Extending the Cell Transmission Model to Multiple Lanes and Lane-Changing*, Section 5, supports local R1 allocation, weighted receiving and a proportional rival, not (3)'s continuous source or the complete compulsory server. Primary manuscript.

Jin, *Continuous formulations and analytical properties of the link transmission model*, published 2015, primary preprint arXiv:1405.7080, Sections 3.1–3.3, supports empty-link queue, vacancy, travel and storage relations from boundary histories. Its general invariant junction result is not attributed to the weighted node here. Primary preprint.

Yperman, Logghe and Immers (2005), *The Link Transmission Model: An Efficient Implementation of the Kinematic Wave Theory in Traffic Networks*, Sections 3–4, supports link/node separation and the need for a stated merge allocation. It does not calibrate this lane-end work factor. Primary paper.

Gate disposition. The original Pass 2 evidence review is complete. A unique, conservative, finite-clearance joined analytical member exists with genuine early service limitation and forced waiting. The user selected T1; this assembled exact contract awaits confirmation before freeze. Applied MP6, observer matching, gradient correctness and independent usefulness remain unestablished. If storage/admission, ordinary-load, initial-symmetry or response conditions fail, reopen the smallest affected G2/G3/G4 contract; do not clip mass, add queues or change laws silently. No implementation, installation, traffic experiment or next review has begun.